



Q1. What is the expected lead time for surface-mount device (SMD) capacitors? Please help us with this information so we can arrange for the circuit fabrication.

Sasank K.

Through E-mail

A1. The lead time is different for different manufacturers. If you need large quantities of a component, well above minimum order quantity (MOQ), it is recommended that you directly discuss with the manufacturer. The advantage is that they will supply the components in standard package, safe from static and humidity. The manufacturer may direct you to the authorised distributor who will provide you the best quote and lead time depending on the availability of that component in the warehouse. The unauthorised distributors break the standard package and supply smaller quantities at higher prices, but the components come with a reduced shelf life and the risk of damage due to static electricity.

For smaller quantities, you can search your parts on the below-mentioned websites and you can see the price and lead time mentioned right there:

1. Element14 (in.element14.com)
2. RS Components (www.rs-components.com)
3. DigiKey (www.digikey.com)
4. Avnet Express (<https://avnetexpress.avnet.com>)

If the component is a common one, you can buy it even from the local markets, such as Lajpat Rai Market in Delhi, with no lead time. But try to buy the components in standard packaging.

Q2. What is the 'remote sense' feature mentioned in the datasheet for various equipment such

as digital power supplies. And how load regulation is different from line regulation?

Karunesh

Through E-mail

A2. This is a very useful feature that ensures that the exact expected voltage is delivered to the load. Some amount of output voltage is always dropped in the cables and the voltage to the load is always less than the meter indication. This feature is implemented in the power supplies by using another pair of port to read back the voltage at the load and auto-correct the output of the power supply accordingly.

Load regulation. This specification defines the consistency in the output even if the load changes. If the load impedance changes, ideally the set parameters should not change significantly. Load regulation is mentioned in the datasheets as $\pm(\% \text{ Set Value} + \text{offset value})$.

Line regulation. This specification defines the consistency in the output even if the input supply or frequency to the power supply varies. This is also represented in the same way as load regulation.

Q3. I have successfully been able to burn the OS in Raspberry Pi with Win32 Disk Imager using Windows system and boot it. Please tell me how to do it with Linux system.

Dinesh Rai

Through E-mail

A3. Insert the SD card in the SD card slot of your Linux system. Issue command 'ls /dev/sd*' at the terminal prompt and see what name has been given to the SD card

as shown in the figure below. Look for the highest order like sdb, sdc, etc. The result will be something as mentioned below:

```
/dev/sda /dev/sda2 /dev/sda4
/dev/sda6 /dev/sdb /dev/sdb2
/dev/sda1 /dev/sda3 /dev/sda5 /dev
/sda7 /dev/sdb1
```

In the above example, SD card has been identified as /dev/sdb2. To be sure, just unplug and re-plug the card a few times and notice the changes.

Now use 'cd' command and go to the folder where you have extracted downloaded image of the Raspberry Pi OS.

```
sunil@sunil-HP-430-Notebook-PC: ~
sunil@sunil-HP-430-Notebook-PC:~$ ls /dev/sd*
/dev/sda /dev/sda2 /dev/sda4 /dev/sda6 /dev/sdb
/dev/sda1 /dev/sda3 /dev/sda5 /dev/sda7
sunil@sunil-HP-430-Notebook-PC:~$ ls /dev/sd*
/dev/sda /dev/sda2 /dev/sda4 /dev/sda6 /dev/sdb /dev/sdb2
/dev/sda1 /dev/sda3 /dev/sda5 /dev/sda7 /dev/sdb1
sunil@sunil-HP-430-Notebook-PC:~$
```

SD card detection

```
$ cd Desktop/Raspberry-Pi/OS
```

Desktop/Raspberry-Pi/OS is the directory where we extracted the image.

This image file now needs to be written on the SD card to make it the OS for Raspberry Pi. At the terminal prompt, issue the following command:

```
$ sudo dd if=2012-09-18-wheezy-raspbian.img of=/dev/sdb2 bs=4M
```

Supply the sudo password when prompted. This will use dd, the powerful image creator program of Linux, and transfer the image to the SD card. Other than wheezy, Arch Linux will also get installed in the same way. However, for XBMC, the Internet connection is required while transferring the image onto the SD card.

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/L, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender